Number Sequences & Patterns

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

A) Easy: Find the next two terms: 3, 6, 9, 12, __, __.
Method & answer: Add 3 each time. Next terms: 15, 18.

B) Standard: Find the next two terms: 2, 5, 10, 17, __, __.
Method & answer: The differences increase by 2 each time (3, 5, 7...). Next: 17+9=26, 26+11=37.

C) Advanced: Find the nth term rule for 4, 7, 10, 13, ... and the 10th term.
Method & answer: Common difference 3, first term 4, so nth term = 3n+1. 10th term = 3×10+1 = 31.

Easy / Standard — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. Find the next two terms: 2, 4, 6, 8, __, __.

2. Find the next two terms: 5, 10, 15, 20, __, __.

3. Find the next two terms: 1, 3, 5, 7, __, __.

4. Find the missing term: 4, 8, 12, __, 20. A) 14 B) 15 C) 16 D) 18

5. Find the next two terms: 100, 90, 80, __, __.

6. Select ALL the sequences below that count up in steps of 3: A) 3,6,9,12 B) 2,4,6,8 C) 15,18,21,24 D) 5,10,15,20 E) 21,24,27,30

7. Which is the rule for the sequence 5, 10, 15, 20...? A) add 5 B) add 10 C) multiply by 5 D) subtract 5

8. Find the missing term: 2, 4, __, 8, 10.

9. Find the next two terms: 50, 45, 40, __, __.

10. Find the next two terms: 7, 14, 21, __, __.

11. Find the missing term: 10, 20, __, 40, 50.

12. Find the next two terms: 1, 4, 7, 10, __, __.

13. Which is the rule for the sequence 20, 18, 16, 14...? A) subtract 1 B) subtract 2 C) subtract 3 D) add 2

14. Select ALL the sequences below that step up in 6s: A) 6,12,18,24 B) 5,10,15,20 C) 9,15,21,27 D) 2,7,12,17 E) 10,16,22,28

15. Find the missing term: 9, __, 15, 18, 21.

Standard — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. Find the next two terms: 2, 4, 8, 16, __, __.

2. Find the next two terms: 1, 4, 9, 16, __, __.

3. Find the next two terms: 1, 3, 6,

6. The nth term of a sequence is 4n − 1. Find the 5th term.

7. Find the next two terms: 0.5, 1, 1.5, 2, __, __.

8. Select ALL the sequences below

11. The nth term of a sequence is 5n. Which of these is the 8th term? A) 35 B) 40 C) 45 D) 13

12. Find the next two terms: -5, -2, 1, 4, __, __.

13. Find the next term: 100, 50, 25,

10, __, __.

that decrease by the same amount each time: A) 20,17,14,11 B) 30,25,20,15 C) 12,10,9,7 D) 40,32,24,16 E) 50,45,40,36

__. A) 15 B) 12 C) 12.5 D) 10

4. Select ALL the sequences below where each term doubles the one before: A) 3,6,12,24 B) 2,4,6,8 C) 5,10,20,40 D) 1,2,4,9 E) 7,14,28,56

9. Find the missing term: 3, 9, 27, __, 243.

14. Find the rule and the next term: 2, 6, 18, 54, __.

5. The nth term of a sequence is 2n + 3. Which of these is the 3rd term? A) 7 B) 8 C) 9 D) 10

10. Find the next two terms: 1, 1, 2, 3, 5, __, __.

15. Find the missing term: 4, 8, 16, __, 64.

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the nth term rule for the sequence 5, 8, 11, 14, …

6. A pattern of matchstick squares uses 4 sticks for 1 square, 7 for 2 squares, 10 for 3 squares. How many sticks for 8 squares?

11. Find the next term: 1, 2, 4, 7, 11, __.

2. Using the nth term rule 3n+2, is 47 a term in the sequence? A) Yes, when n = 15 B) Yes, when n = 16 C) No, 47 is never reached D) No, but 44 is

7. Find the missing terms: 1, __, 9, __, 25.

12. A sequence begins 3, 7, 11, 15… Find the first term greater than 100.

3. Find the nth term rule for the sequence 2, 5, 8, 11, … and find the 20th term.

8. Find the nth term rule for 7, 12, 17, 22, …

13. Find the rule and the 6th term: 1, 3, 7, 15, 31, __.

4. A sequence starts at 100 and decreases by 7 each time. What is the 10th term?

9. Using the rule 5n+2, does the sequence ever equal 100? A) Yes, when n=19 B) Yes, when n=20 C) No, because (100−2)/5 is not a whole number D) No, because 100 is too large

14. Two sequences are given by 2n + 1 and 3n − 2. For what value of n are they equal? A) n = 2 B) n = 3 C) n = 4 D) n = 5

5. The nth term of a sequence is $n^2 + 1$. Select ALL of these that are terms in the sequence: A) 2 B) 5 C) 8 D) 10 E) 17

10. A sequence has nth term $2n^2$. Select ALL of these that are terms: A) 8 B) 20 C) 32 D) 40 E) 48

15. A sequence of triangular numbers is 1, 3, 6, 10, 15… What is the 10th triangular number?